

Applications of Triangle Congruence and More Triangle Theorems

Applications of Triangle Congruence

In a previous class we stated the following definition.

Original: If corresponding parts of polygons are congruent, then the polygons are congruent.

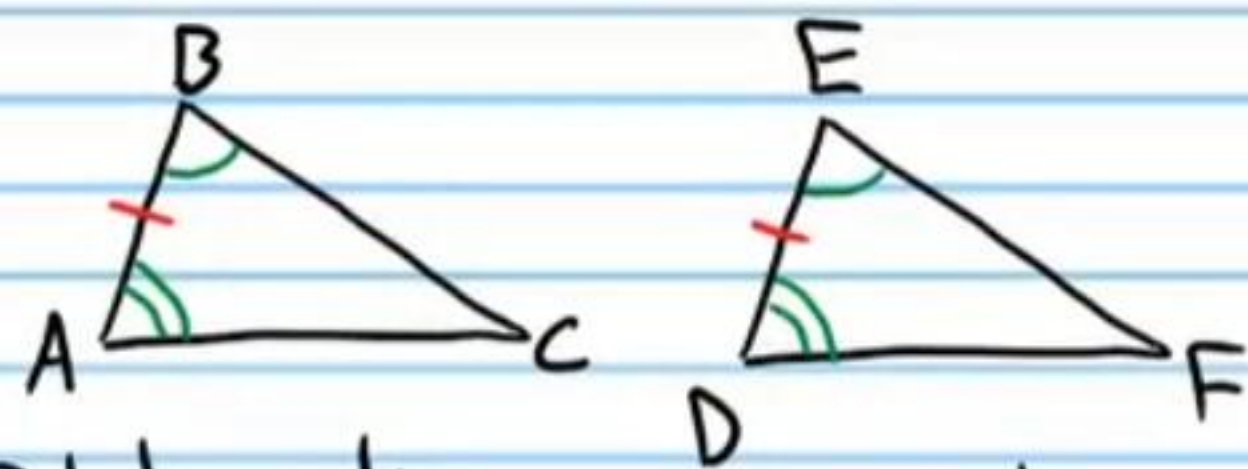
The converse of a definition is always true.

Converse: If polygons are congruent, then corresponding parts of the polygons are congruent.

Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

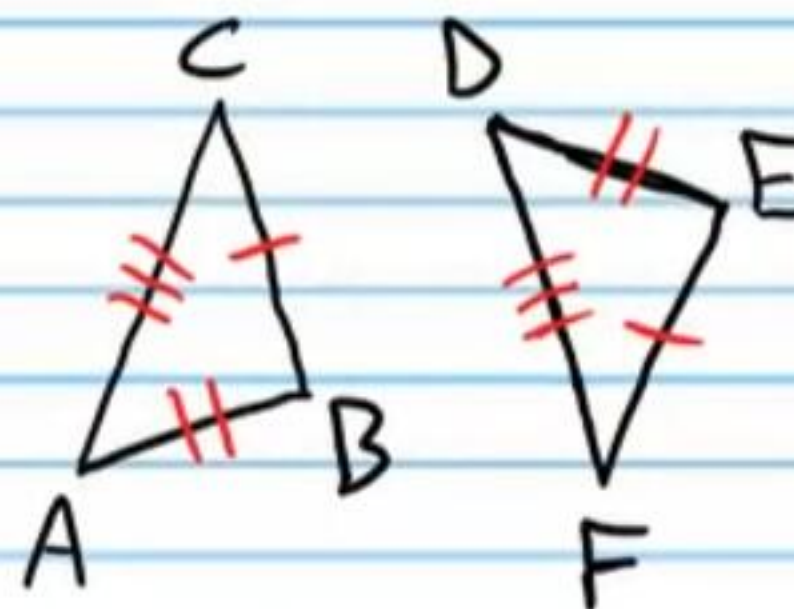
Write two-column proofs to show that the statements below are true.

① If $\angle B \cong \angle E$, $\angle A \cong \angle D$, and $\overline{AB} \cong \overline{DE}$, then $\overline{BC} \cong \overline{EF}$.



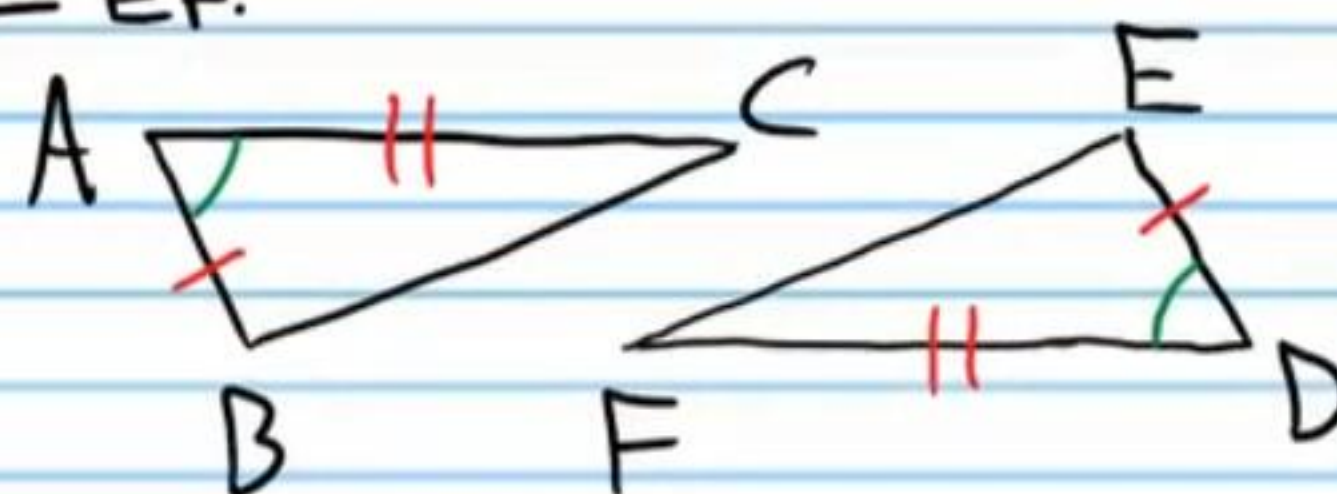
Statement	Reason
I) $\angle B \cong \angle E$ & $\angle A \cong \angle D$.	Given.
II) $\overline{AB} \cong \overline{DE}$	Given.
III) $\triangle ABC \cong \triangle DEF$	ASA
IV) $\overline{BC} \cong \overline{EF}$	CPCTC

② If $\overline{AC} \cong \overline{DF}$, $\overline{AB} \cong \overline{DE}$, and $\overline{BC} \cong \overline{EF}$, then $\angle B \cong \angle E$.



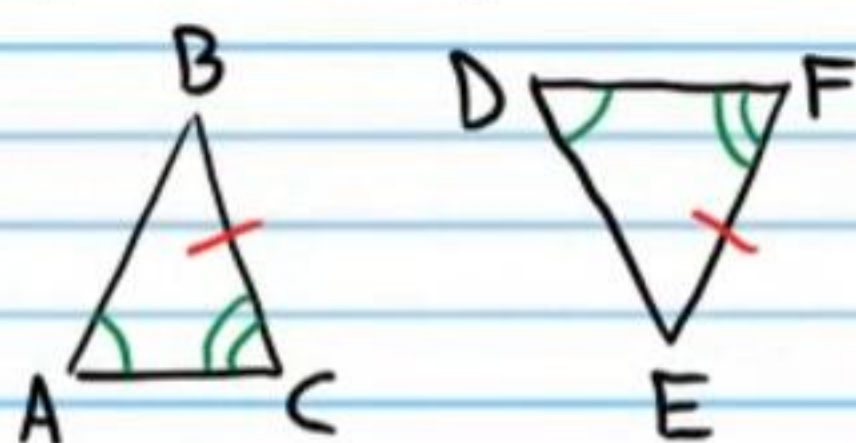
Statement	Reason
I) $\overline{AC} \cong \overline{DF}$ & $\overline{AB} \cong \overline{DE}$ & $\overline{BC} \cong \overline{EF}$.	Given.
II) $\triangle ABC \cong \triangle DEF$	SSS
III) $\angle B \cong \angle E$	CPCTC

③ If $\overline{AB} \cong \overline{DE}$, $\overline{AC} \cong \overline{DF}$, and $\angle A \cong \angle D$ below, then $\overline{BC} \cong \overline{EF}$.



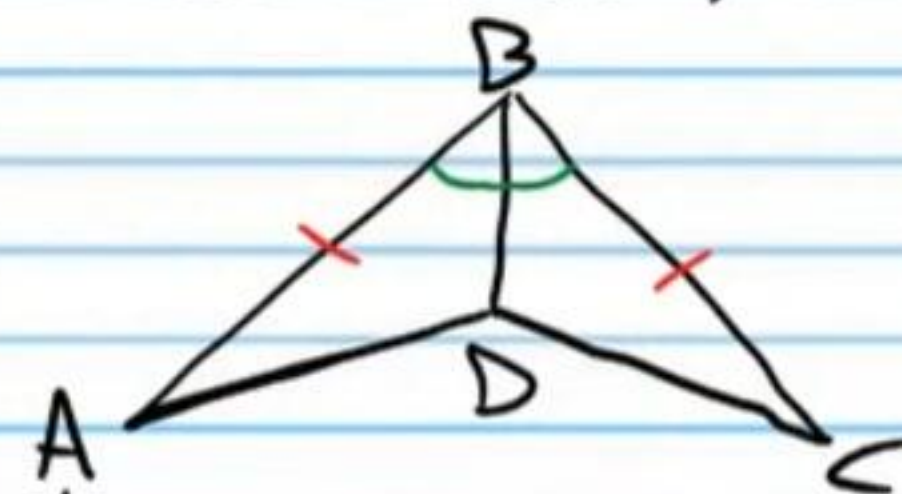
Statement	Reason
I) $\overline{AB} \cong \overline{DE}$ & $\overline{AC} \cong \overline{DF}$ & $\angle A \cong \angle D$	Given.
II) $\triangle ABC \cong \triangle DEF$	SAS
III) $\overline{BC} \cong \overline{EF}$	CPCTC

④ If $\angle C \cong \angle F$, $\angle A \cong \angle D$, and $\overline{BC} \cong \overline{EF}$, then $\overline{AB} \cong \overline{DE}$.



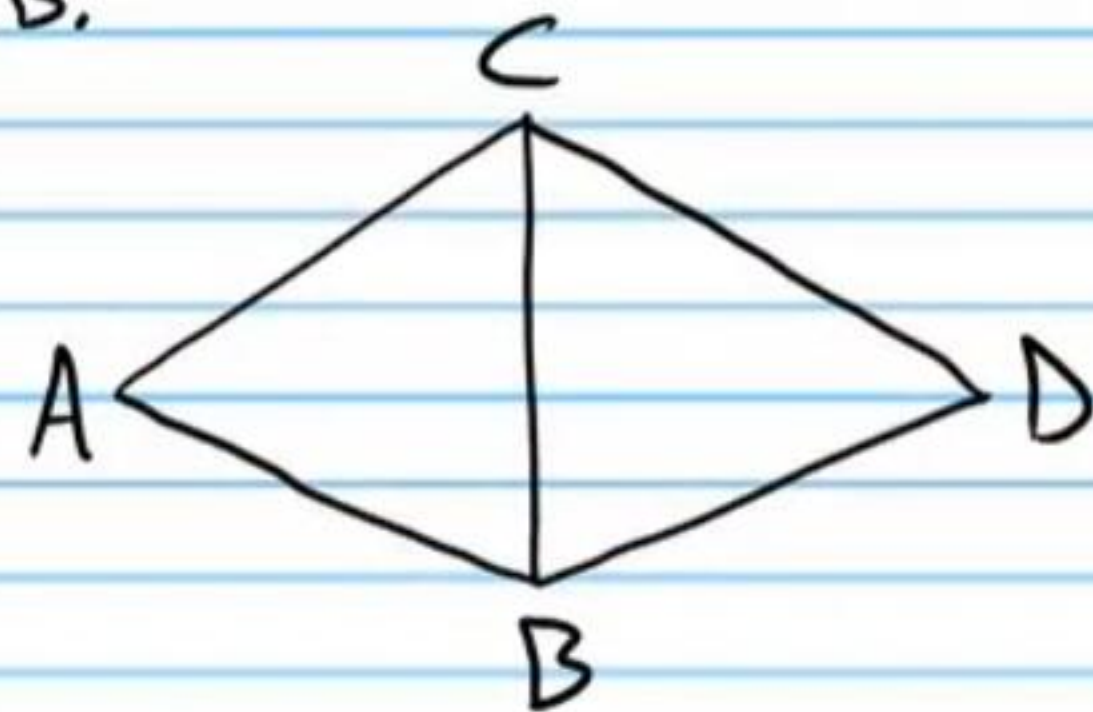
Statement	Reason
I) $\angle C \cong \angle F$ & $\angle A \cong \angle D$ & $\overline{BC} \cong \overline{EF}$	Given.
II) $\triangle ABC \cong \triangle DEF$	AAS
III) $\overline{AB} \cong \overline{DE}$	CPCTC

⑤ If $\angle ABD \cong \angle CBD$ below, and $\overline{AB} \cong \overline{CB}$, then $\angle A \cong \angle C$.



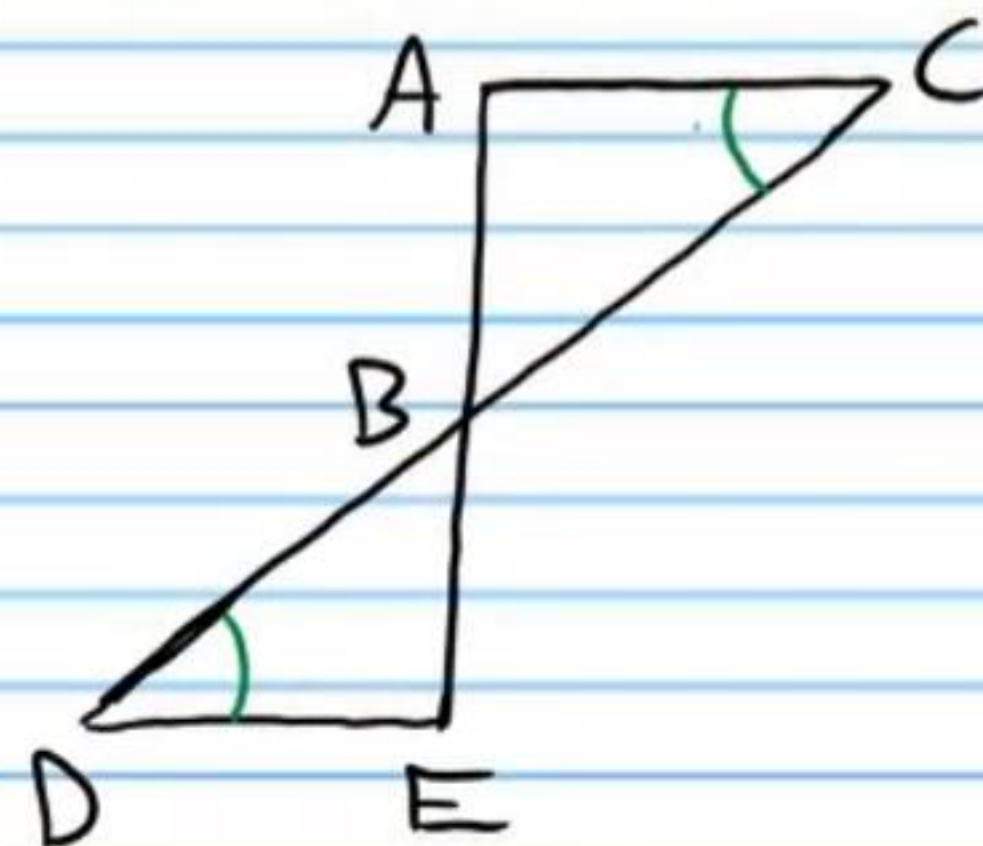
Statement	Reason
I) $\angle ABD \cong \angle CBD$	Given.
II) $\overline{AB} \cong \overline{CB}$	Given.
III) $\overline{BD} \cong \overline{BD}$	Ref Prop of $\cong$
IV) $\triangle ABD \cong \triangle CBD$	SAS
V) $\angle A \cong \angle C$	CPCTC

⑥ If $\overline{AB} \cong \overline{BD}$ and $\overline{BC}$ bisects $\angle ABD$, then $\angle ACB \cong \angle DCB$.



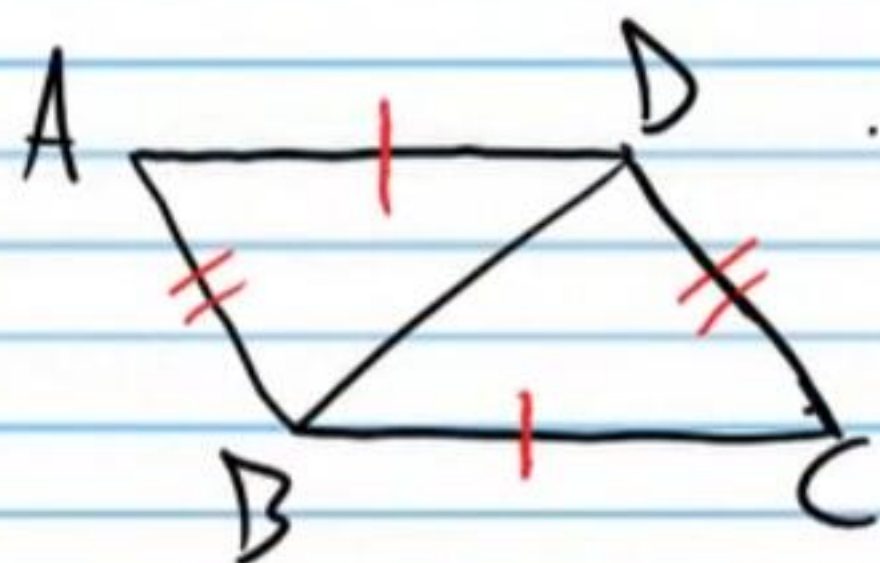
Statement	Reason
I) $\overline{AB} \cong \overline{BD}$	Given.
II) $\overline{BC}$ bisects $\angle ABD$.	Given.
III) $\angle ABC \cong \angle DBC$	Def. of $\angle$ bisector.
IV) $\overline{BC} \cong \overline{BC}$	Ref Prop of $\cong$
V) $\triangle ABC \cong \triangle DCB$	SAS
VI) $\angle ACB \cong \angle DCB$	CPCTC

⑦ If $\angle ACB \cong \angle EDB$ below and B is the midpoint of $\overline{AE}$, then B is also the midpoint of $\overline{CD}$.



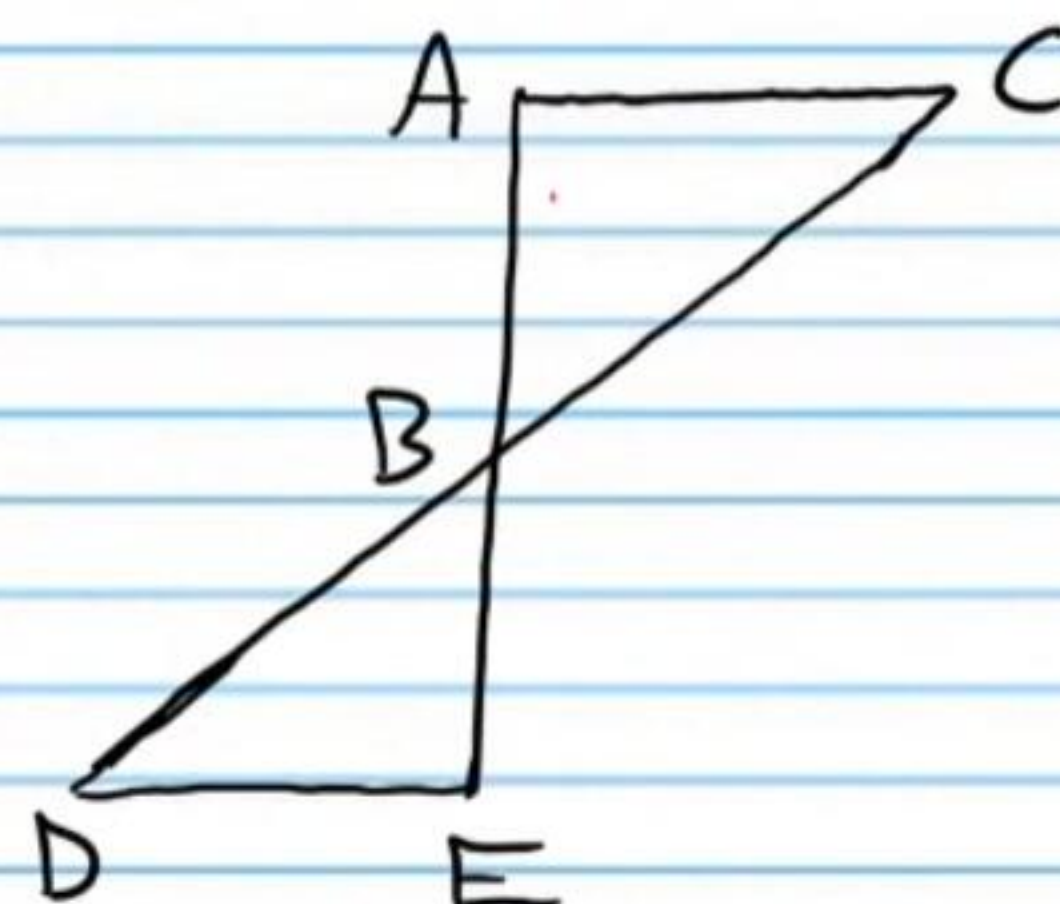
Statement	Reason
I) $\angle ACB \cong \angle EDB$	Given.
II) B is the midpoint of $\overline{AE}$.	Given.
III) $\overline{AB} \cong \overline{BE}$	Def. of midpoint
IV) $\angle ABC \cong \angle EBD$	Vert. $\angle$'s Thm.
V) $\triangle ABC \cong \triangle EBD$	AAS
VI) $\overline{CB} \cong \overline{BD}$	CPCTC
VII) B is the midpoint of $\overline{CD}$.	Def of midpoint

⑧ If $\overline{AD} \cong \overline{BC}$ and $\overline{AB} \cong \overline{CD}$ below, then $\angle A \cong \angle C$.



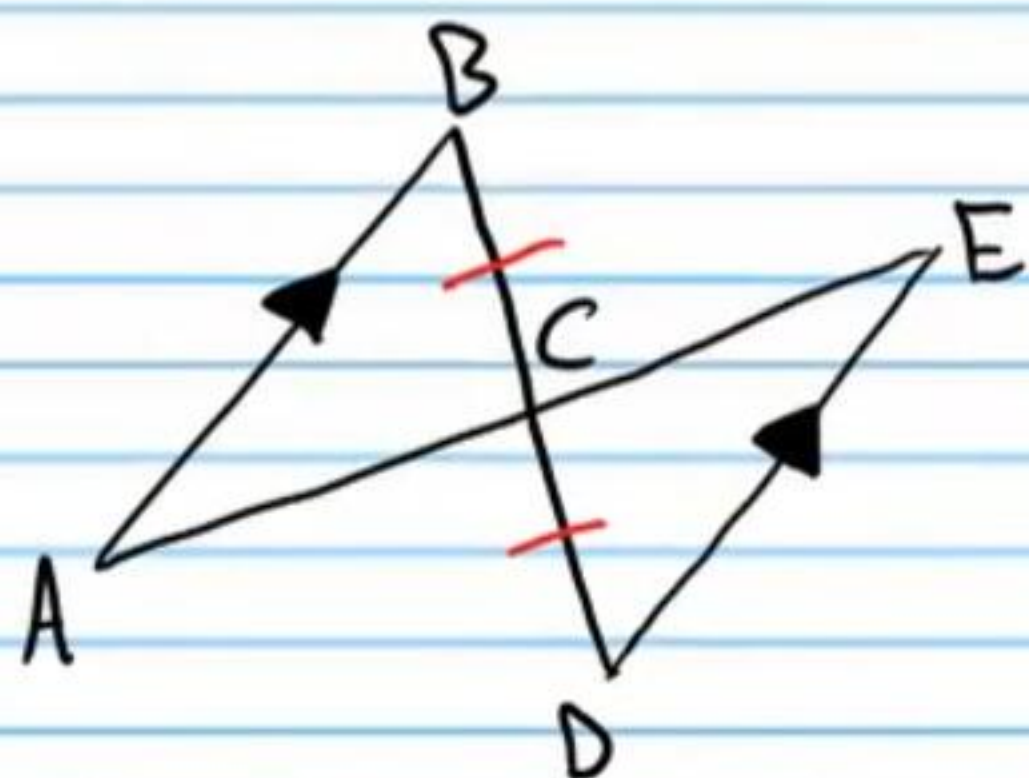
Statement	Reason
I) $\overline{AD} \cong \overline{BC}$ & $\overline{AB} \cong \overline{CD}$	Given.
II) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
III) $\triangle ABD \cong \triangle CDB$	SSS
IV) $\angle A \cong \angle C$	CPCTC

⑨ If B is the midpoint of $\overline{AE}$ and $\overline{DC}$, then $\overline{AC} \cong \overline{ED}$.



Statement	Reason
I) B is the midpoint of $\overline{AE}$ & $\overline{DC}$	Given.
II) $\overline{AB} \cong \overline{BE}$ & $\overline{DB} \cong \overline{BC}$	Def of midpoint.
III) $\angle ABC \cong \angle EBD$	Vert. $\angle$'s Thm.
IV) $\triangle ABC \cong \triangle EBD$	SAS
V) $\overline{AC} \cong \overline{ED}$	CPCTC

⑩ If $\overline{AB} \parallel \overline{ED}$ below, and $\overline{BC} \cong \overline{DC}$, then C is the midpoint of $\overline{AE}$.

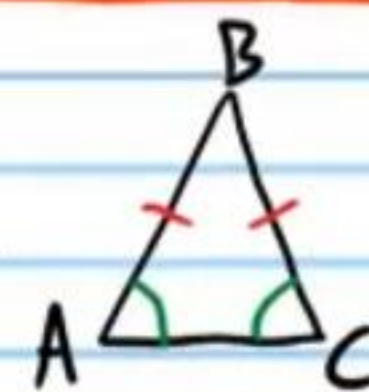


Statement	Reason
I) $\overline{AB} \parallel \overline{ED}$	Given.
II) $\angle ABC \cong \angle EDC$	Alt. Int $\angle$'s Thm
III) $\angle ACB \cong \angle ECD$	Vert. $\angle$'s Thm.
IV) $\overline{BC} \cong \overline{DC}$	Given.
V) $\triangle ABC \cong \triangle EDC$	ASA
VI) $\overline{AC} \cong \overline{CE}$	CPCTC
VII) C is the midpoint of $\overline{AE}$.	Def. of midpoint.

More Triangle Theorems

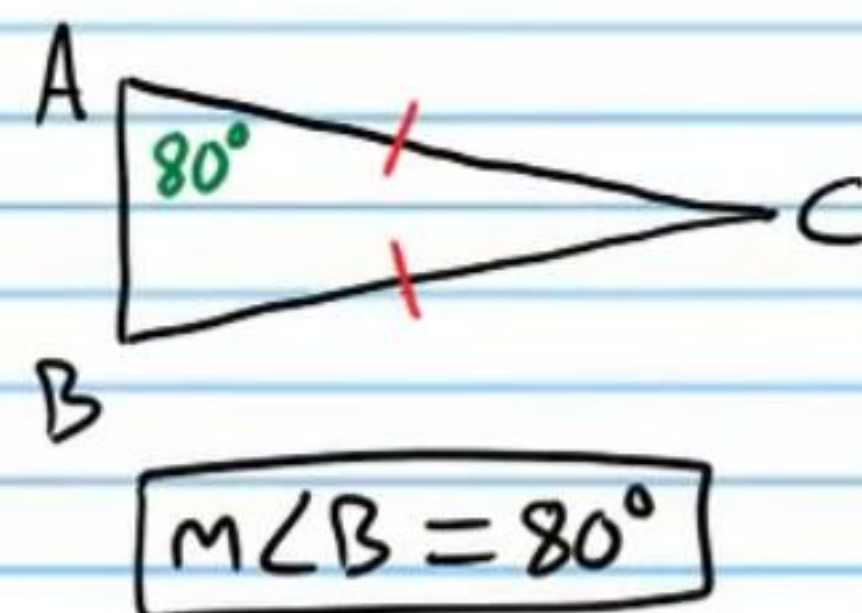
The Isosceles Triangle Theorem: If two sides of a triangle are congruent, then the angles opposite those sides are congruent.

Example: If $\overline{AB} \cong \overline{BC}$, then $\angle A \cong \angle C$.

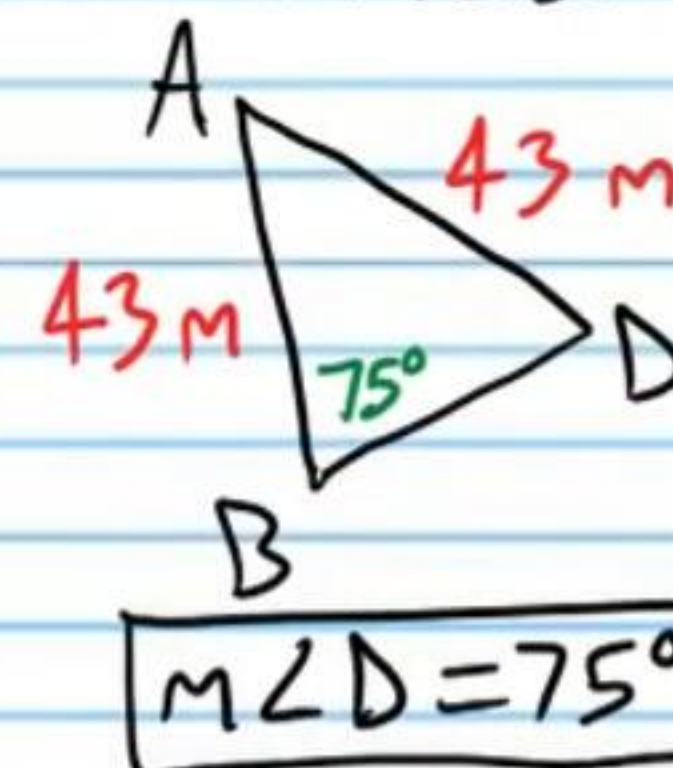


In the following problems, you do not have to write formal proofs.

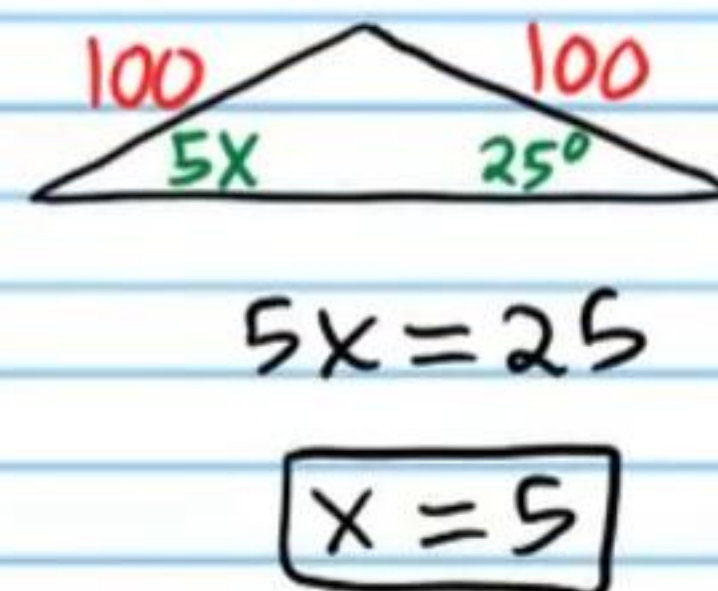
⑪ Find $m\angle B$.



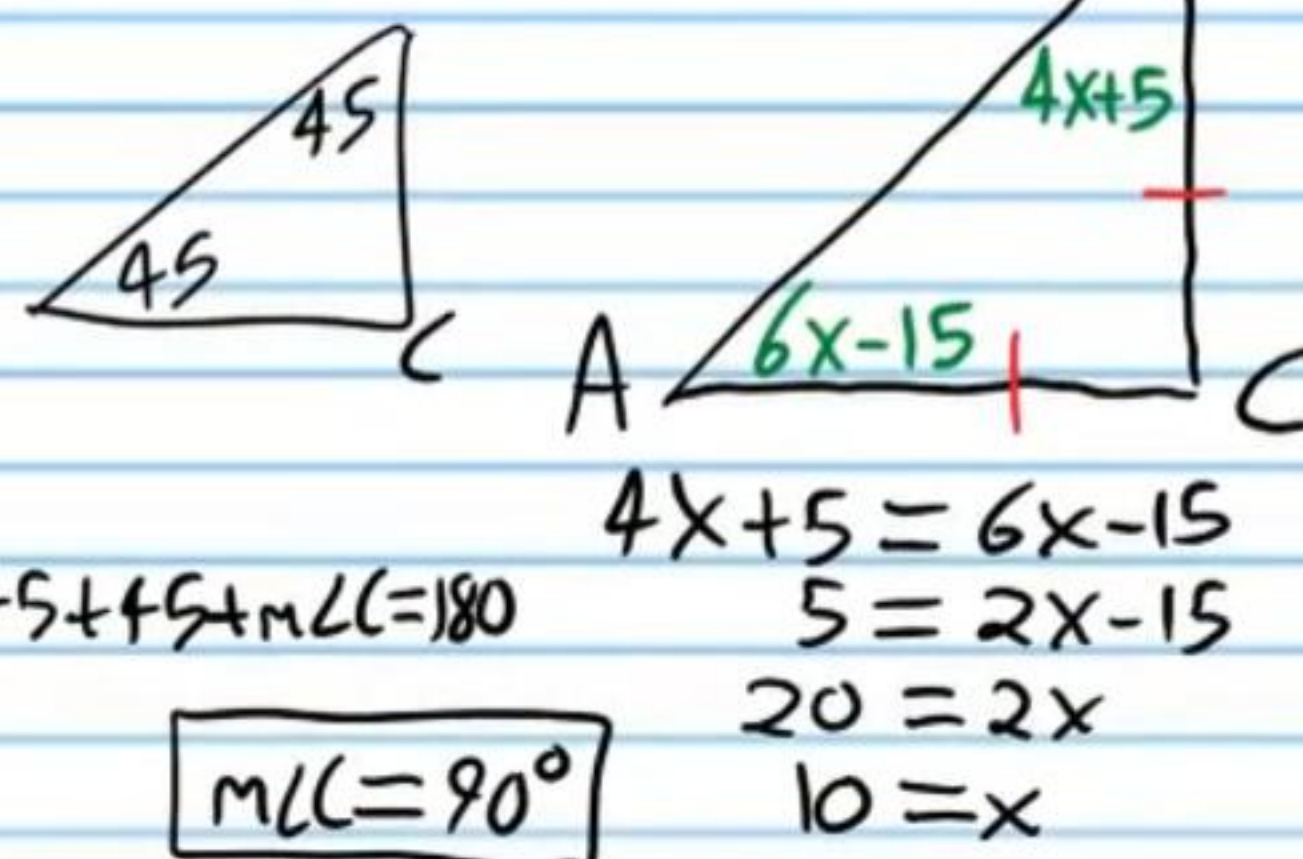
⑫ Find $m\angle D$.



⑬ Find x .



⑭ Find $m\angle C$.

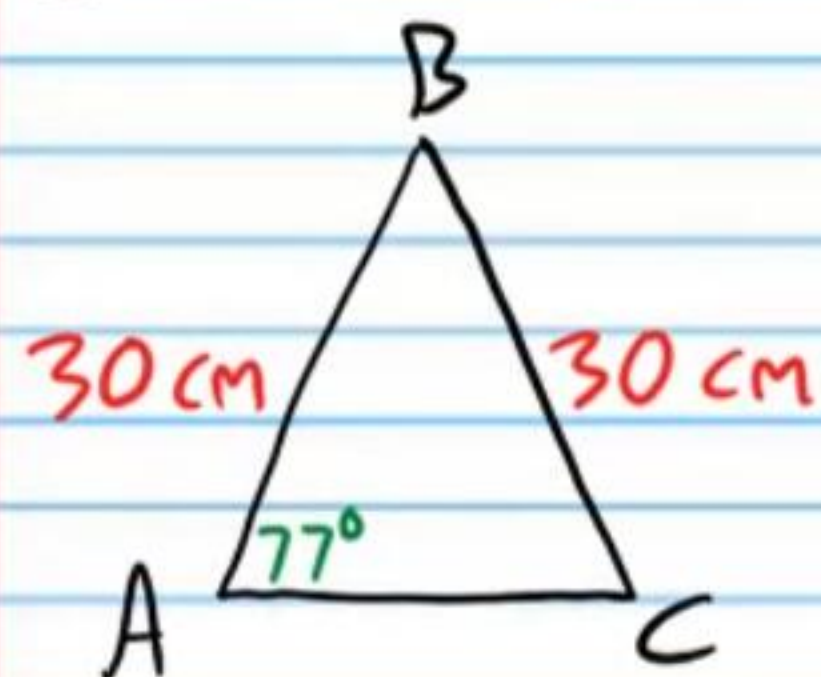


$$45 + 45 + m\angle C = 180$$

$$m\angle C = 90^\circ$$

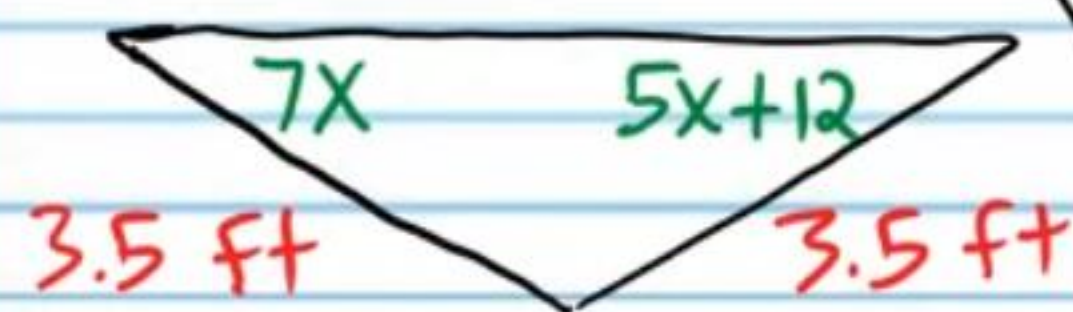
$$\begin{aligned} 4x + 5 &= 6x - 15 \\ 5 &= 2x - 15 \\ 20 &= 2x \\ 10 &= x \end{aligned}$$

⑮ Find $m\angle C$.



$$\boxed{m\angle C = 77^\circ}$$

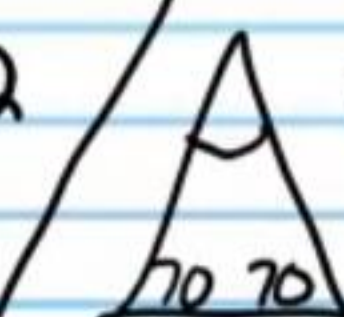
⑰ Find x .



$$7x = 5x + 12$$

$$2x = 12$$

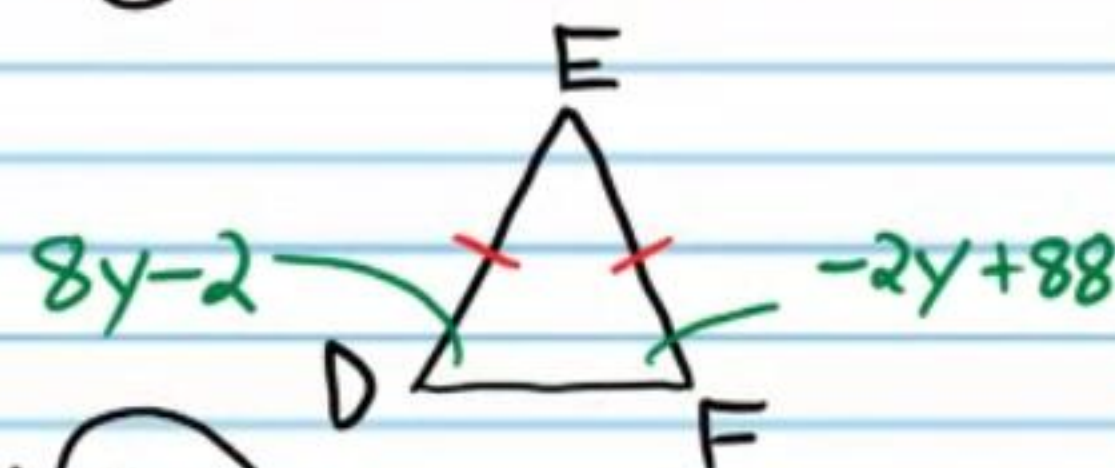
$$\boxed{x = 6}$$



$$70 + 70 + m\angle E = 180$$

$$\boxed{m\angle E = 40^\circ}$$

⑱ Find $m\angle E$.



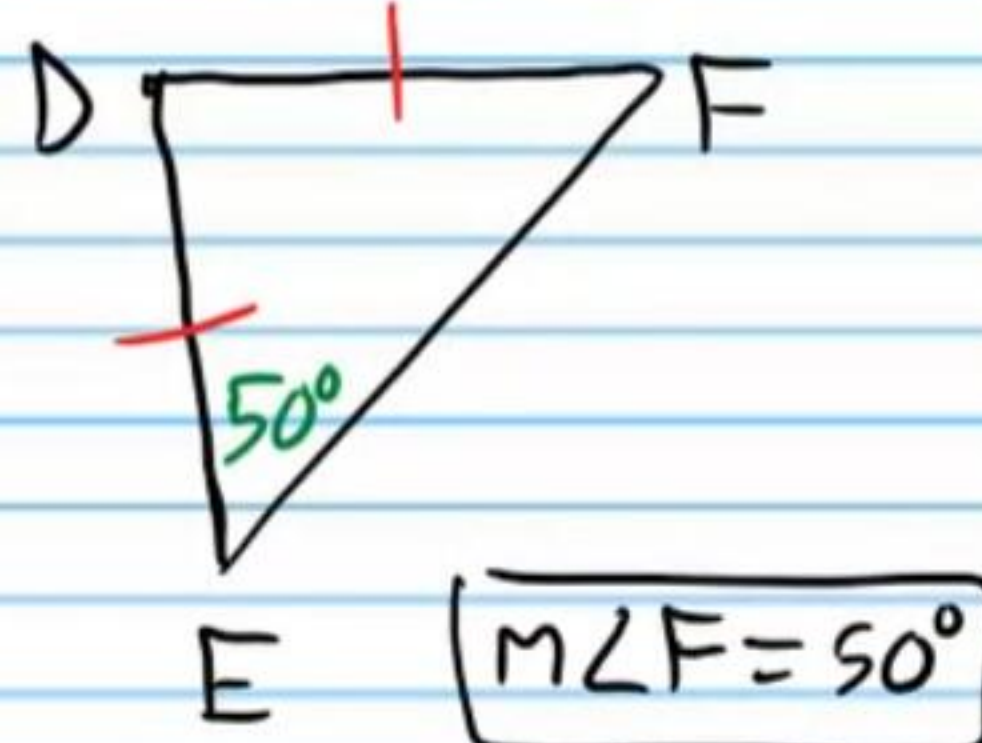
$$8y - 2 = -2y + 88$$

$$10y - 2 = 88$$

$$10y = 90$$

$$y = 9$$

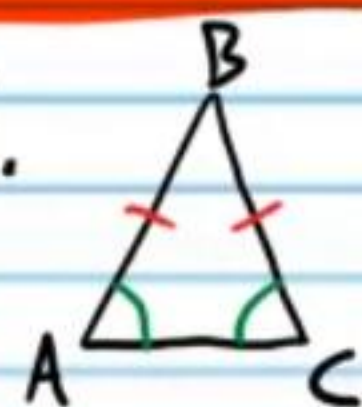
⑯ Find $m\angle F$.



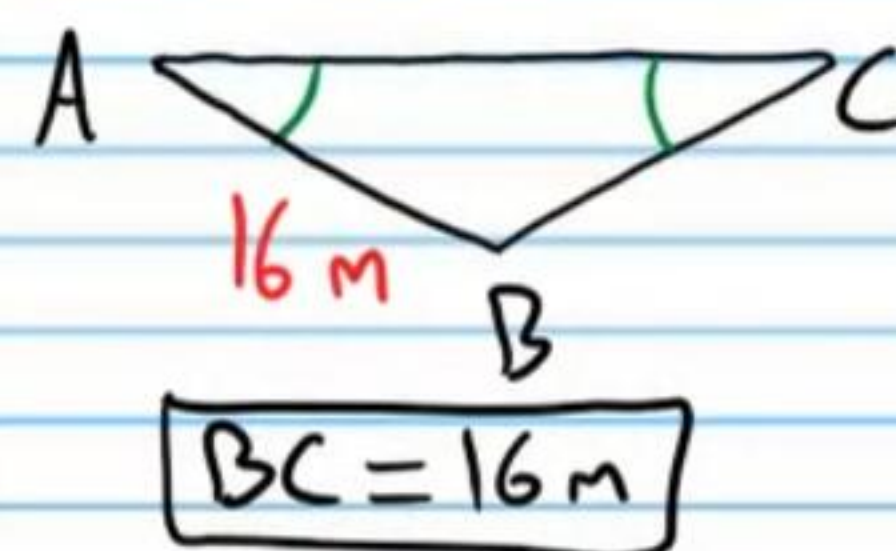
$$\boxed{m\angle F = 50^\circ}$$

The Isosceles Triangle Converse Theorem: If two angles in a triangle are congruent, then the sides opposite those angles must be congruent.

Example: If $\angle A \cong \angle C$, then $\overline{AB} \cong \overline{BC}$.

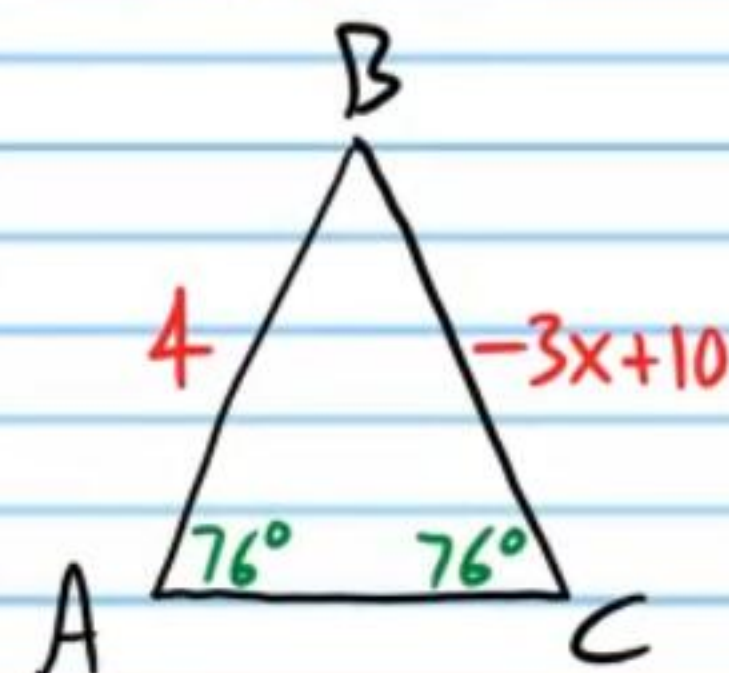


⑲ Find BC .



$$\boxed{BC = 16m}$$

⑳ Find x .

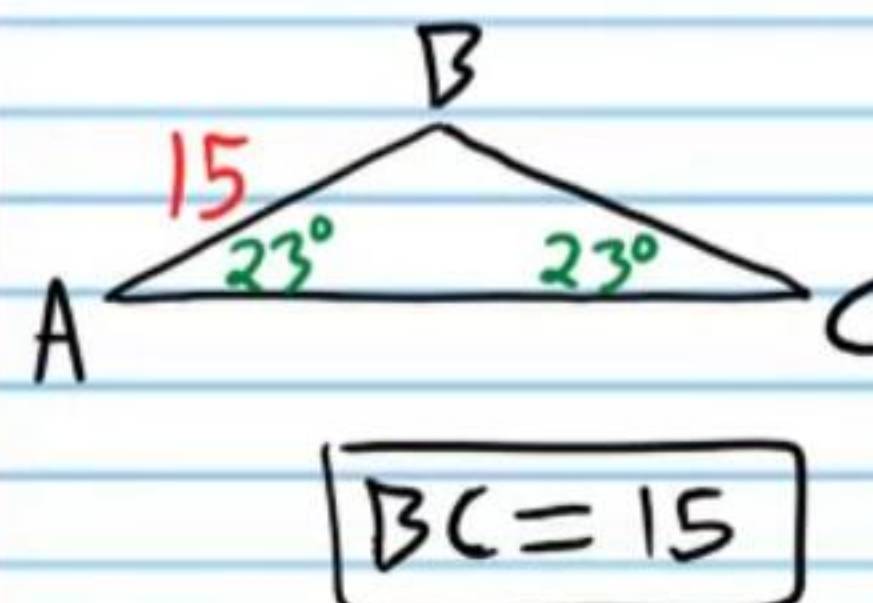


$$4 = -3x + 10$$

$$-6 = -3x$$

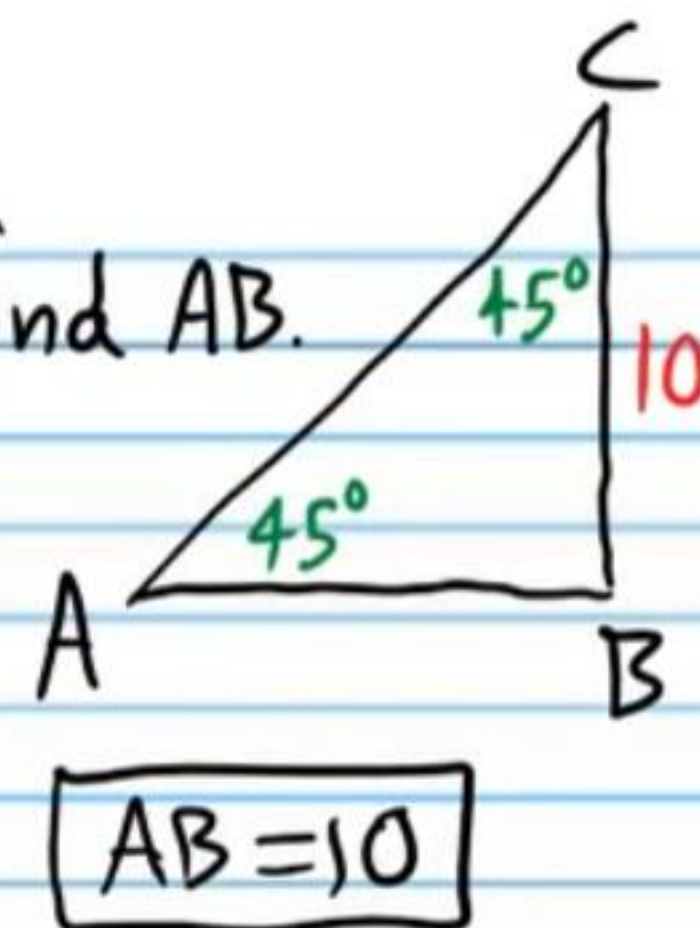
$$\boxed{2 = x}$$

㉓ Find BC .



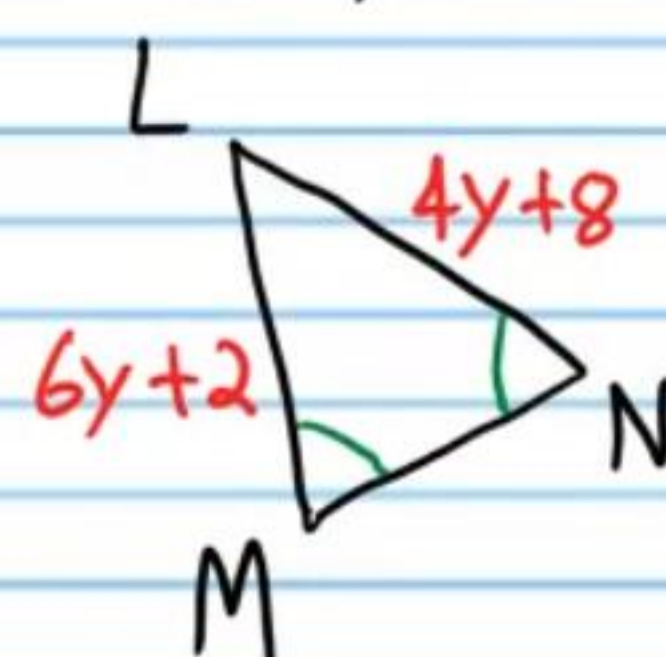
$$\boxed{BC = 15}$$

㉔ Find AB .



$$\boxed{AB = 10}$$

㉔ Find y .



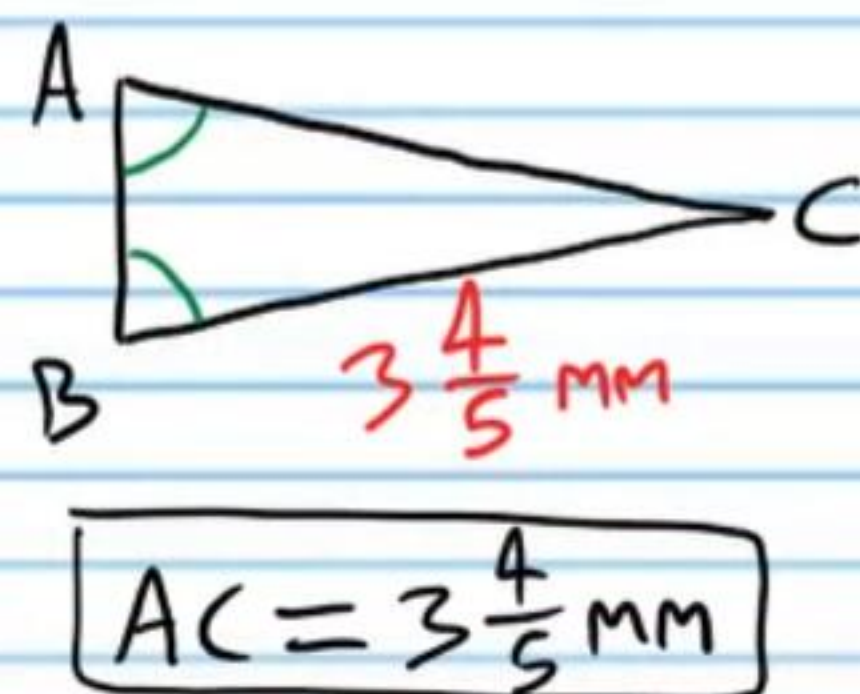
$$6y + 2 = 4y + 8$$

$$2y + 2 = 8$$

$$2y = 6$$

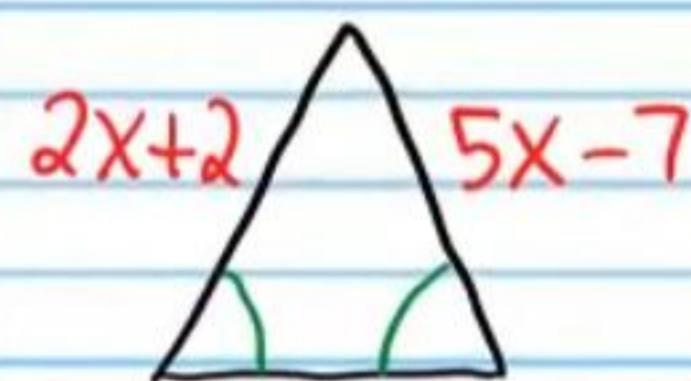
$$\boxed{y = 3}$$

㉔ Find AC .



$$\boxed{AC = 3\frac{4}{5}mm}$$

②5 Find X.



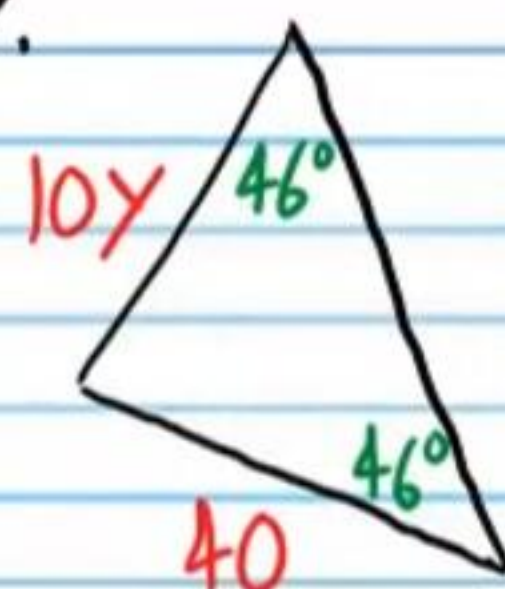
$$2X+2 = 5X-7$$

$$2 = 3X-7$$

$$9 = 3X$$

$$\boxed{3=X}$$

②6 Find y.

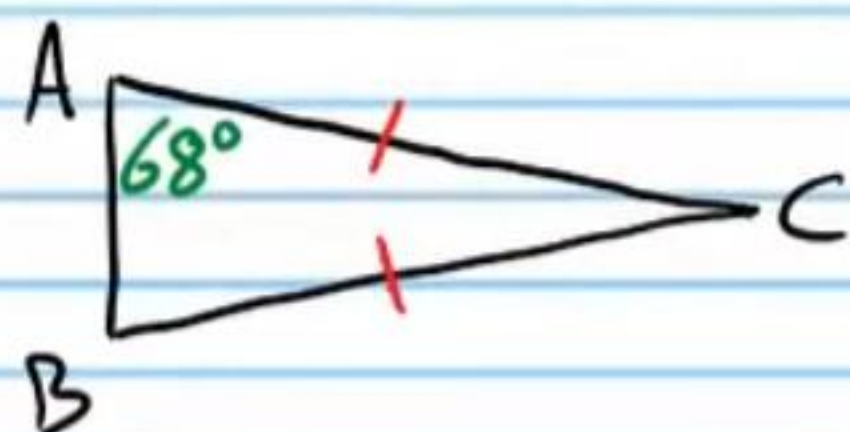


$$10y = 40$$

$$\boxed{y=4}$$

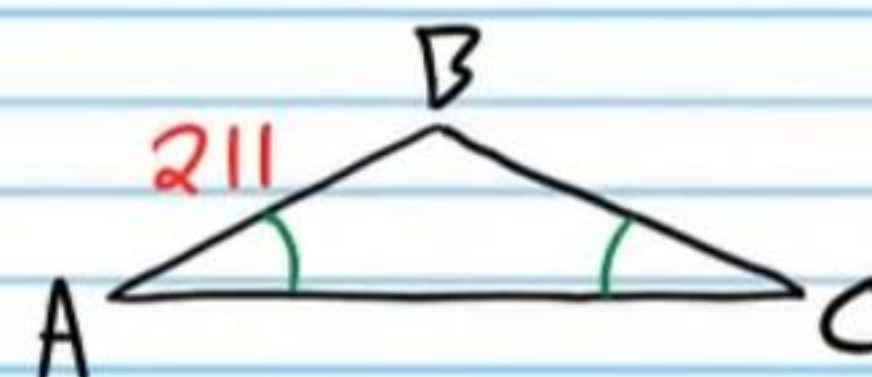
Write two-column proofs to show that the following statements are true.

②7 If $m\angle A = 68^\circ$ and $\overline{AC} \cong \overline{BC}$, then $m\angle B = 68^\circ$.



Statement	Reason
I) $m\angle A = 68^\circ$	Given.
II) $\overline{AC} \cong \overline{BC}$	Given.
III) $\angle A \cong \angle B$	Isosceles Δ Thm
IV) $m\angle A = m\angle B$	Def of $\cong \angle$.
V) $m\angle B = 68^\circ$	Trans. Prop of =.

②8 If $AB = 211$ and $\angle A \cong \angle C$, then $BC = 211$.



Statement	Reason
I) $AB = 211$	Given.
II) $\angle A \cong \angle C$	Given.
III) $\overline{AB} \cong \overline{BC}$	Isosceles Δ Conv. Thm.
IV) $AB = BC$	Def. of $\cong$ seg
V) $BC = 211$	Trans. Prop of =.

The Equilateral Triangle Theorem: If a triangle is equilateral, then it is equiangular.

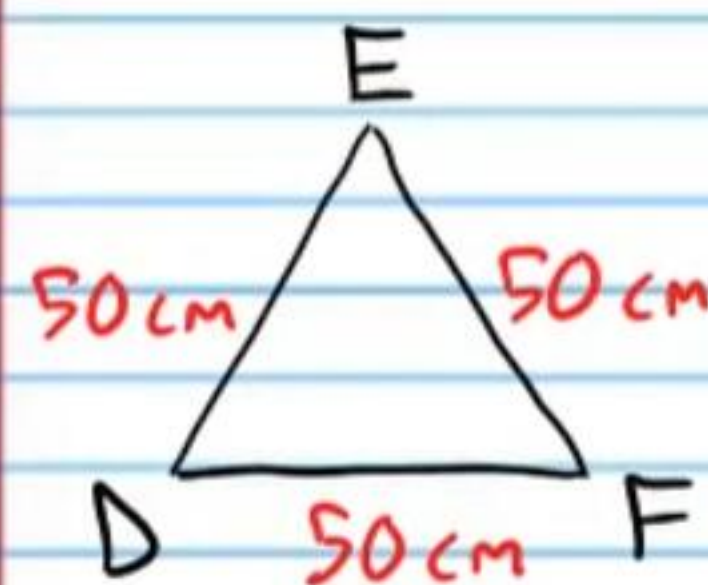
The Equiangular Triangle Theorem: If a triangle is equiangular, then the measure of every interior angle is 60° .

29 Find $m\angle E$.



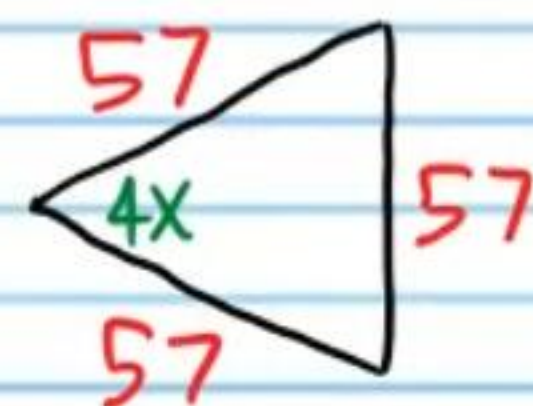
$$m\angle E = 60^\circ$$

31 Find $m\angle D$.



$$m\angle D = 60^\circ$$

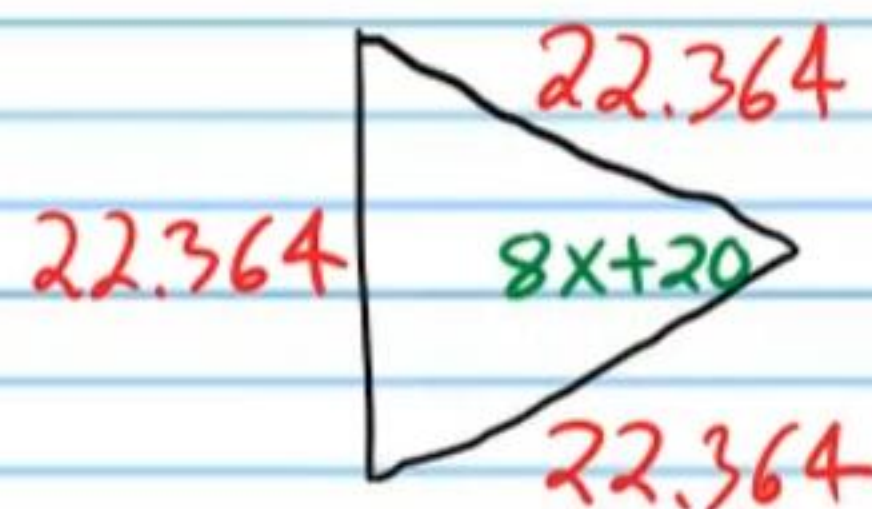
30 Find x .



$$4x = 60$$

$$x = 15$$

32 Find x .



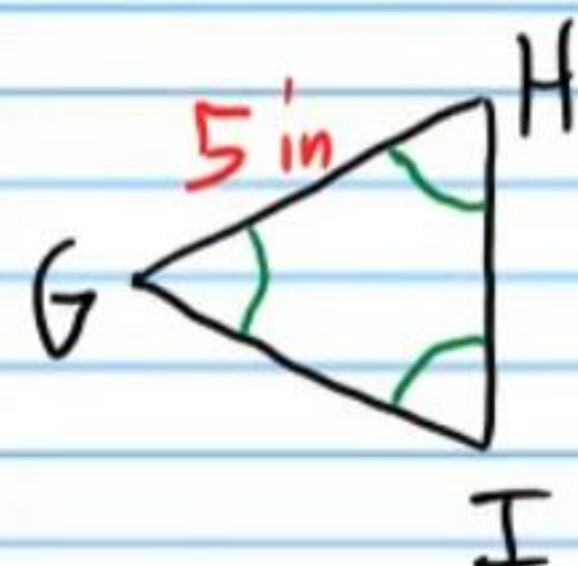
$$8x + 20 = 60$$

$$8x = 40$$

$$x = 5$$

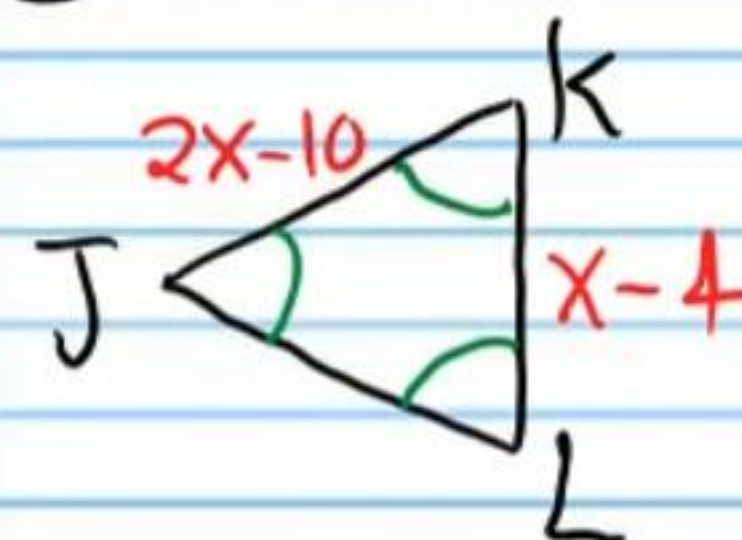
The Equilateral Triangle Converse Theorem: If a triangle is equiangular, then it is equilateral.

33 Find HI .



$$HI = 5 \text{ in}$$

35 Find JL .



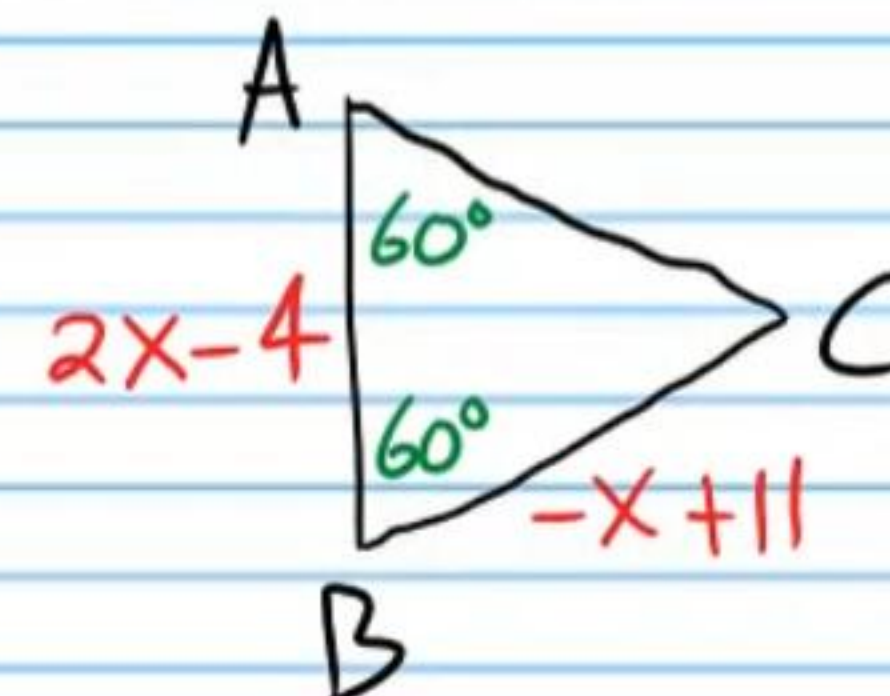
$$2x - 10 = x - 4$$

$$x - 10 = -4$$

$$x = 6$$

$$JL = JK = 2(6) - 10 = 2$$

34 Find x .



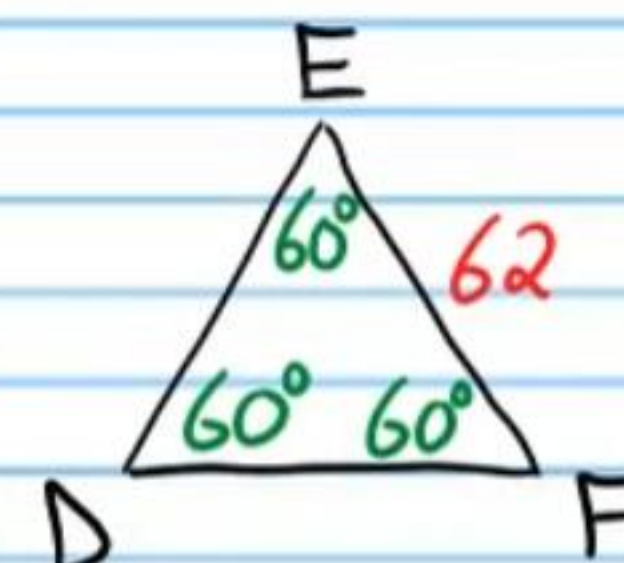
$$2x - 4 = -x + 11$$

$$3x - 4 = 11$$

$$3x = 15$$

$$x = 5$$

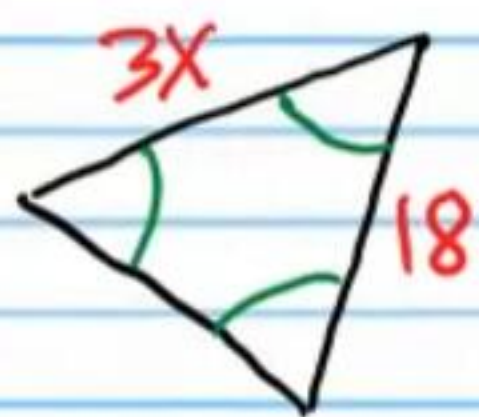
36 Find ED and DF .



$$ED = 62$$

$$DF = 62$$

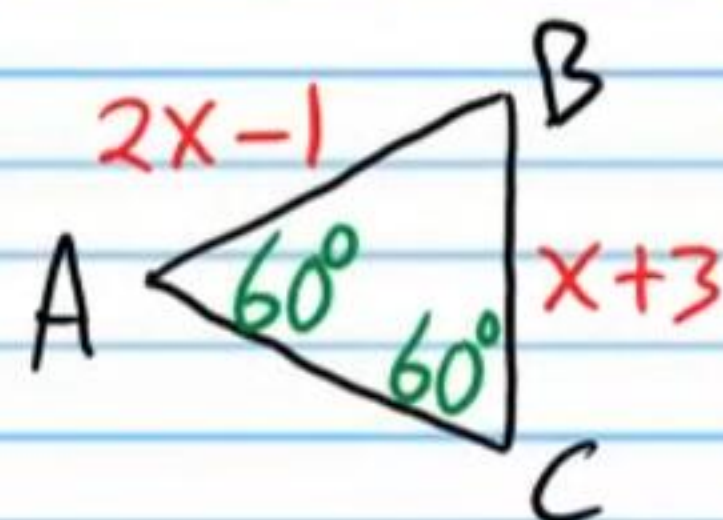
③⑦ Find X .



$$3X = 18$$

$$\boxed{X = 6}$$

③⑧ Find BC .



$$2X - 1 = X + 3$$

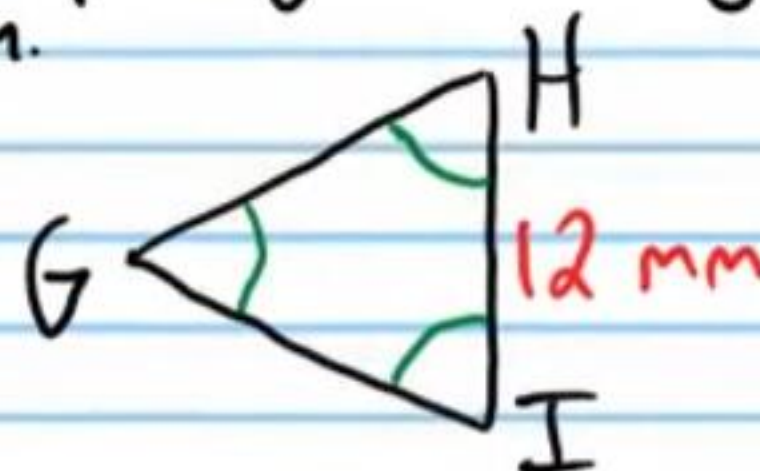
$$X - 1 = 3$$

$$X = 4$$

$$BC = 4 + 3 = \boxed{7}$$

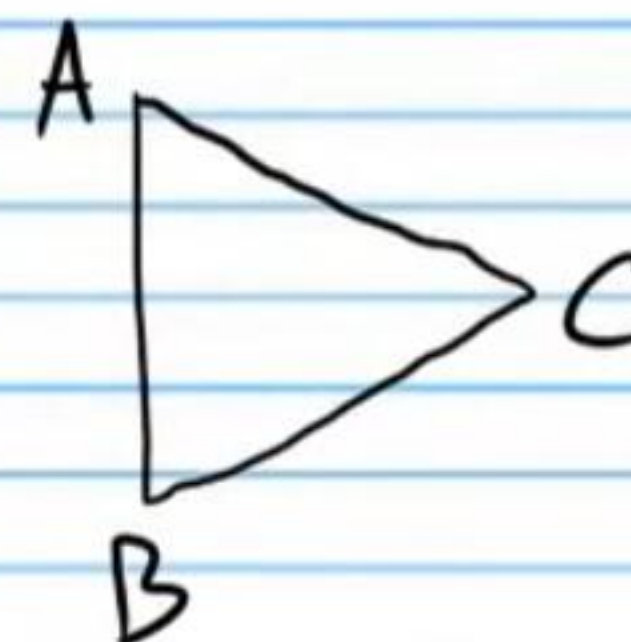
Write two-column proofs to show that the following statements are true.

③⑨ If $\triangle GHI$ below is an equiangular triangle, and $HI = 12\text{ mm}$, then $GI = 12\text{ mm}$.



Statement	Reason
I) $\triangle GHI$ is equiangular.	Given.
II) $\triangle GHI$ is equilateral	Equilateral Δ Conv Thm.
III) $HI = GI$	Def. of equilateral Δ .
IV) $HI = 12\text{ mm}$	Given.
V) $GI = 12\text{ mm}$	Trans. Prop of =.

④⑩ If $AB = AC = BC$ below, then $m\angle C = 60^\circ$.



Statement	Reason
I) $AB = AC = BC$	Given.
II) $\overline{AB} \cong \overline{AC} \cong \overline{BC}$	Def. of $\cong$ Seg.
III) $\triangle ABC$ is equilateral.	Def. of equilateral Δ .
IV) $\triangle ABC$ is equiangular.	Equilateral Δ Thm.
V) $m\angle C = 60^\circ$	Equiangular Δ Thm.

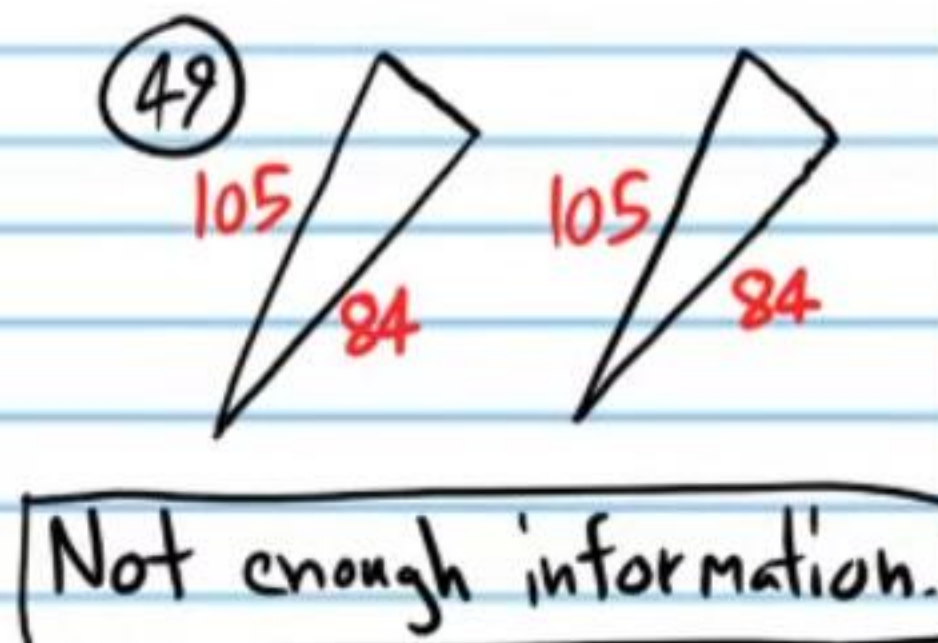
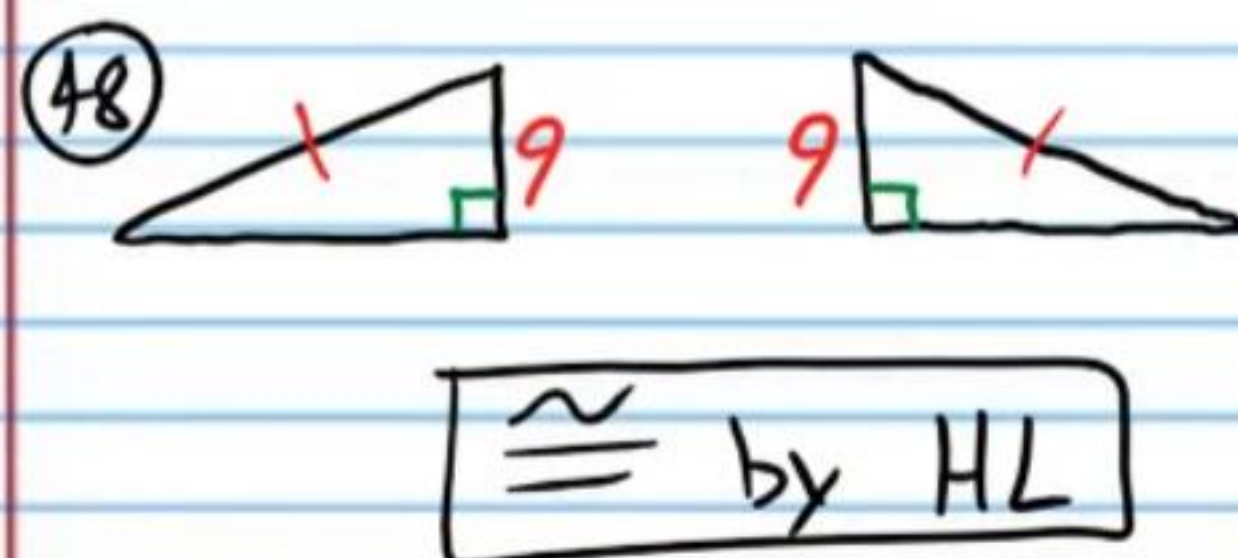
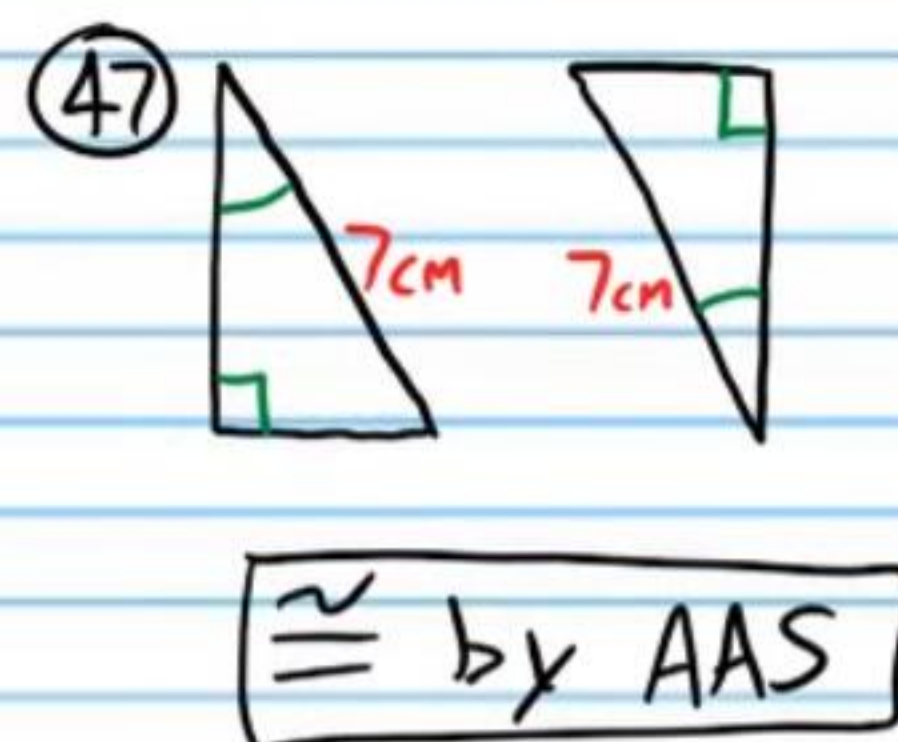
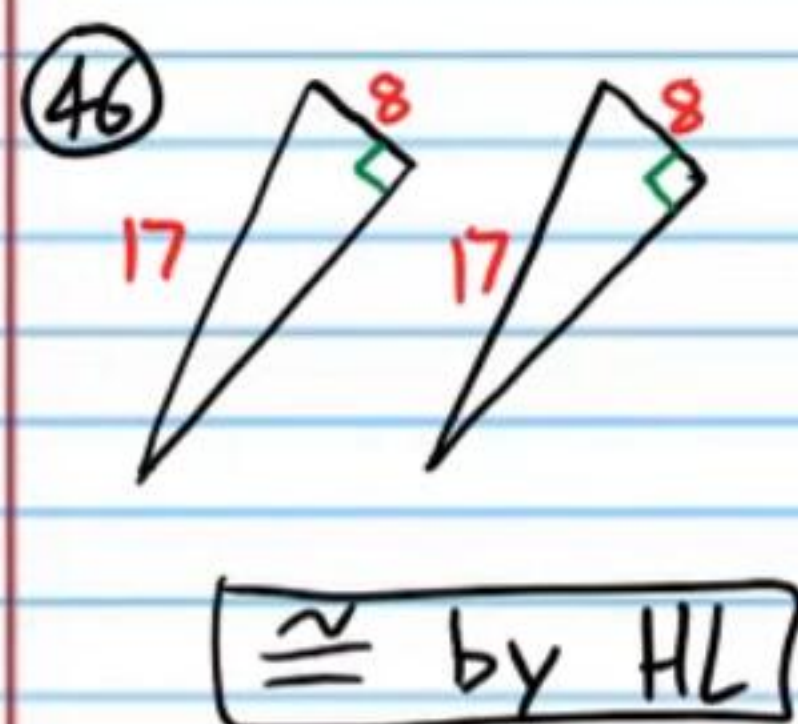
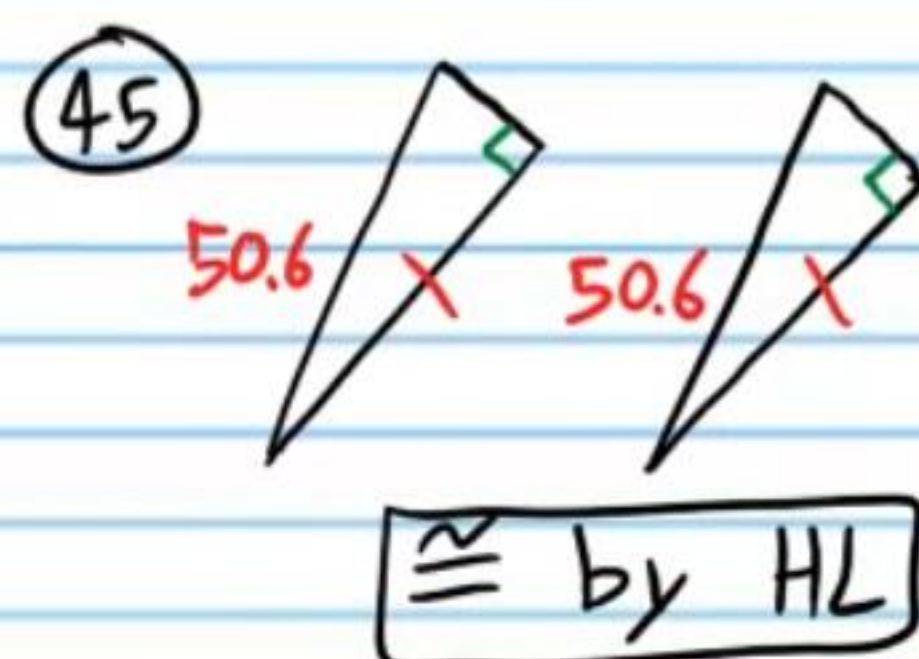
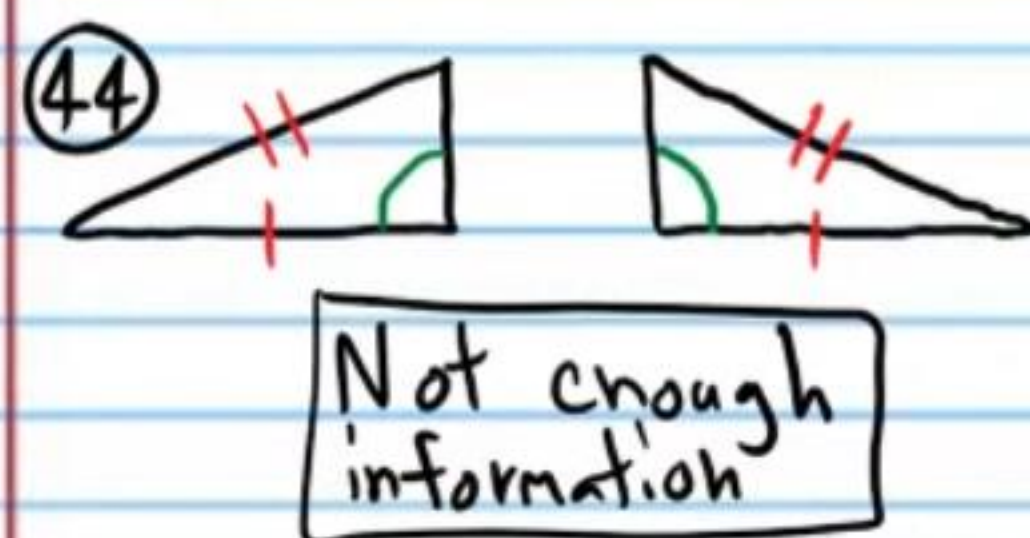
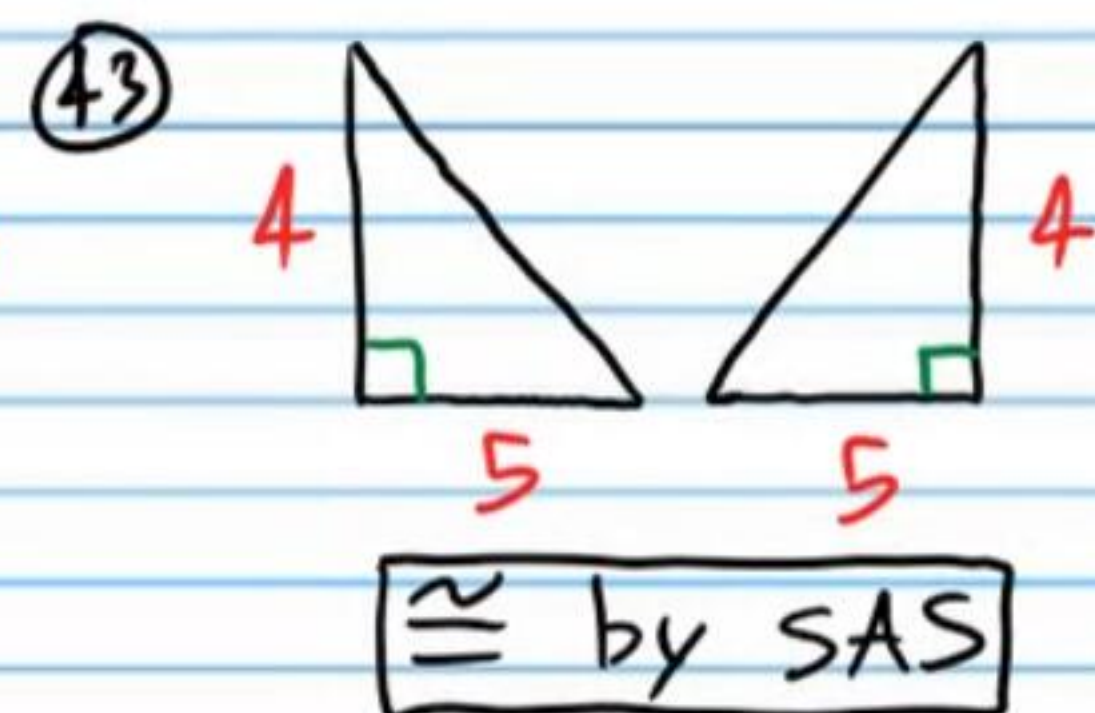
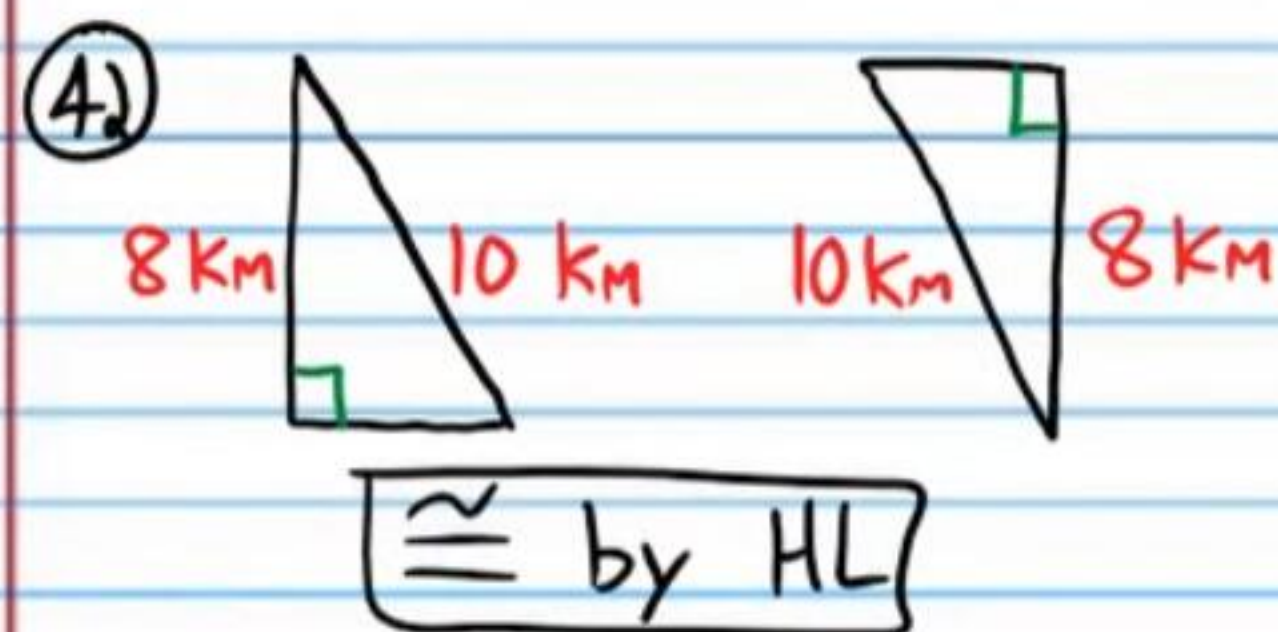
The Hypotenuse-Leg Congruence Theorem: If a hypotenuse and leg of one right triangle are congruent to the hypotenuse and leg of another right triangle, then the triangles are congruent.

Determine if the triangles below are congruent. If so, state the postulate or theorem used.

④⑪

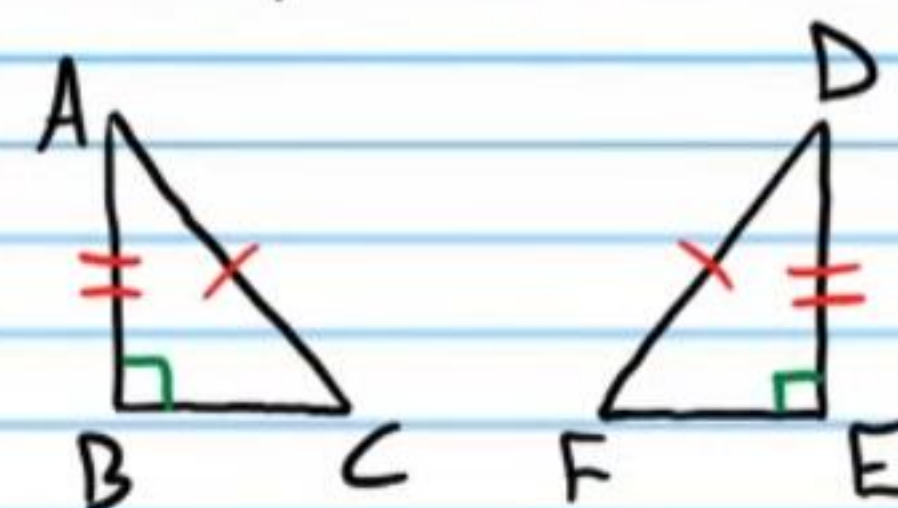


$\cong$ by HL



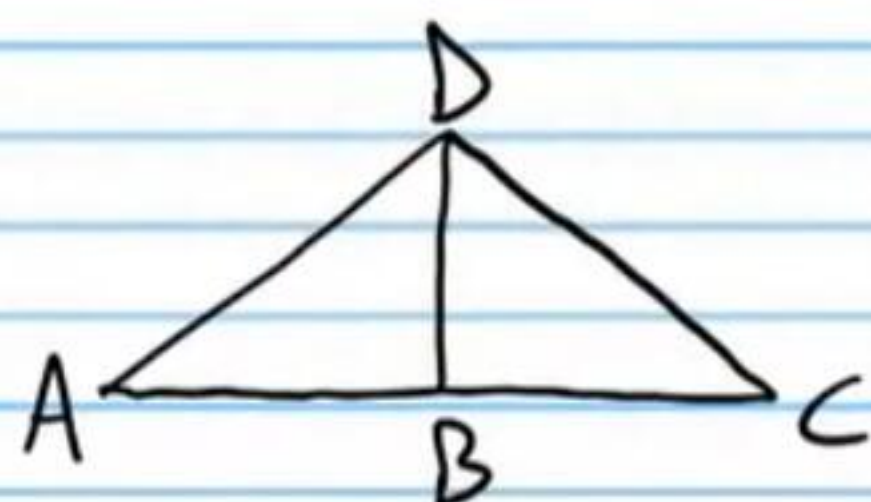
Use the HL Theorem in two-column proofs to show that following statements are true.

50) If $\triangle ABC$ and $\triangle DEF$ are right triangles, and $\overline{AB} \cong \overline{DE}$ and $\overline{AC} \cong \overline{DF}$, then $\angle F \cong \angle C$.



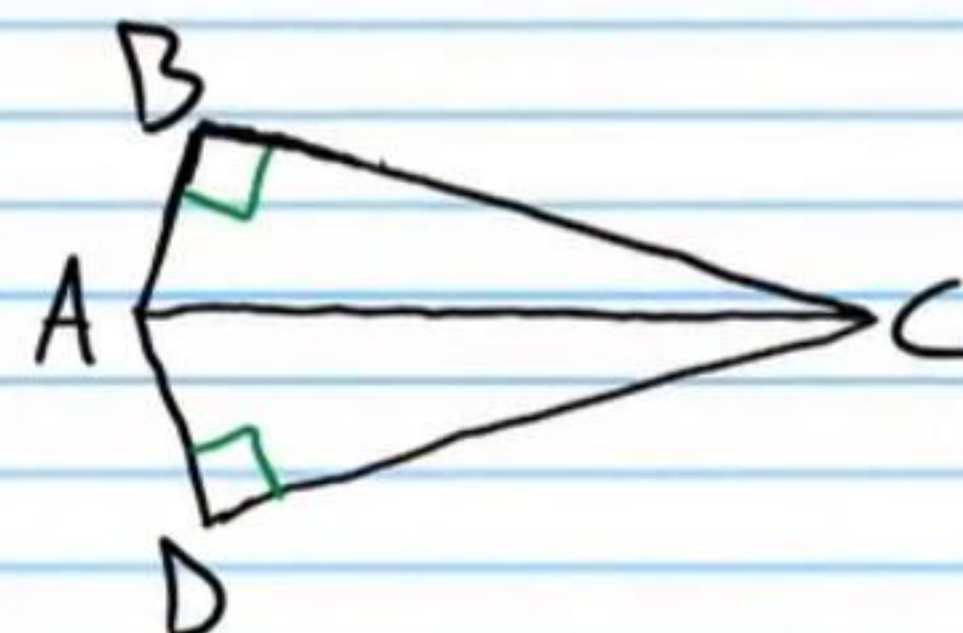
Statement	Reason
I) $\triangle ABC$ & $\triangle DEF$ are Rt $\triangle$'s.	Given.
II) $\overline{AB} \cong \overline{DE}$ & $\overline{AC} \cong \overline{DF}$	Given.
III) $\triangle ABC \cong \triangle DEF$	HL
IV) $\angle F \cong \angle C$	CPCTC

⑤ If $\overline{AC} \perp \overline{BD}$ and $\overline{AD} \cong \overline{DC}$, then $\angle ADB \cong \angle CDB$.



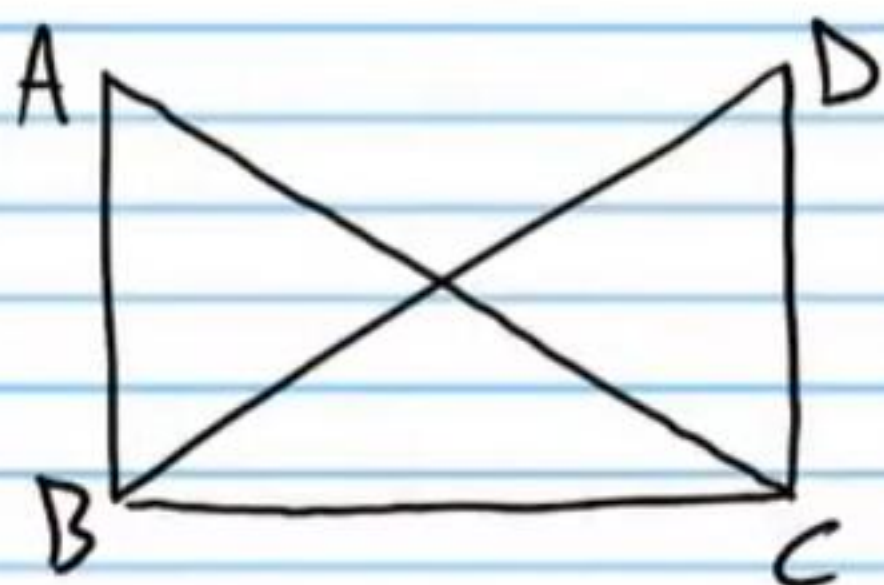
Statement	Reason
I) $\overline{AC} \perp \overline{BD}$	Given.
II) $\angle ABD$ & $\angle CBD$ are rt $\angle$'s	Def of $\perp$
III) $\overline{AD} \cong \overline{DC}$	Given.
IV) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
V) $\triangle ABD \cong \triangle CBD$	HL
VI) $\angle ADB \cong \angle CDB$	CPCTC

⑤② If $\angle B$ and $\angle D$ are right angles, and $\overline{BC} \cong \overline{DC}$, then $\triangle ABC \cong \triangle ADC$.



Statement	Reason
I) $\angle B$ & $\angle D$ are rt $\angle$'s.	Given.
II) $\overline{BC} \cong \overline{DC}$	Given.
III) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$.
IV) $\triangle ABC \cong \triangle ADC$	HL

⑤③ If $\overline{AB} \perp \overline{BC}$, $\overline{DC} \perp \overline{BC}$, and $\overline{AC} \cong \overline{DB}$, then $\angle A \cong \angle D$.



Statement	Reason
I) $\overline{AB} \perp \overline{BC}$ & $\overline{DC} \perp \overline{BC}$	Given.
II) $\angle ABC$ & $\angle BCD$ are rt $\angle$'s	Def. of $\perp$
III) $\overline{AC} \cong \overline{DB}$	Given.
IV) $\overline{BC} \cong \overline{BC}$	Ref. Prop of $\cong$.
V) $\triangle ABC \cong \triangle DCB$	HL
VI) $\angle A \cong \angle D$	CPCTC